Toronto Metropolitan University

Metropolitan Undergraduate Engineering Society

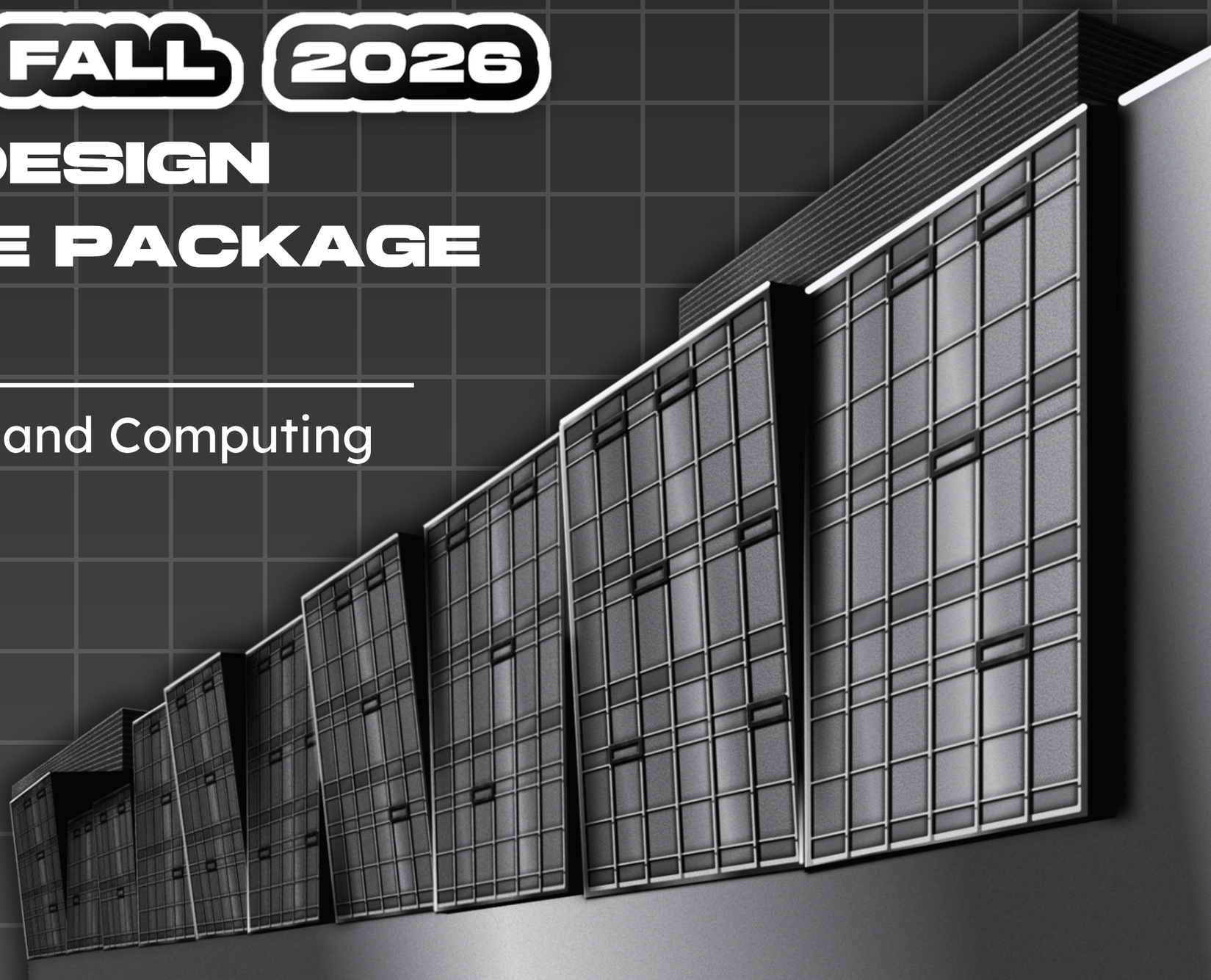
TORONTO METROPOLITAN UNIVERSITY ENGINEERING COMPETITION

FALL 2026

INNOVATIVE DESIGN PRE-ADVANCE PACKAGE

NOVEMBER 7 & 8, 2026

George Vari Engineering and Computing Centre



1. COMPETITION CHALLENGE

The City of Toronto is seeking an innovative engineering proposal for an Urban Food Security District in Scarborough. The theme for this year is Feeding the Future: Designing Smart Infrastructure for Urban Food Security

Teams are tasked with transforming an abandoned industrial site into a functional food security district capable of producing, preserving, storing, processing, and distributing food while remaining resilient to environmental conditions and supply-chain disruptions.

The proposed district should improve food accessibility, reduce food waste, and demonstrate practical engineering solutions that balance sustainability, innovation, and economic feasibility.

2. PROJECT INFORMATION

- Location: Scarborough, Toronto (Birchmount Road and Sheppard Avenue East area)
- Available Land: 40 acres
- Maximum Construction Budget: $50 million CAD
- Minimum Project Lifespan: 20 years
- Team Size: 5–6 competitors
- Competition Date: Sunday, November 8, 2026

For the purposes of this competition, teams may treat the 40 acre site as a hypothetical redevelopment area within the specified Scarborough location. Unless additional site information is provided by the Competition Coordinators, teams may make reasonable assumptions regarding site dimensions, access points, and existing infrastructure. All significant assumptions must be clearly identified and justified.

3. REQUIRED FACILITIES

The proposed Urban Food Security District must include all five required facilities or operational areas. Teams may organize these functions into separate buildings, shared buildings, or integrated infrastructure where appropriate, provided that each required function is clearly identified, adequately supported, and justified through engineering design considerations.

1. Farm / Greenhouse

A designated area for food production, including appropriate growing methods, water management, and supporting infrastructure.

2. Waste Recovery

A system for collecting and processing organic waste into reusable resources, such as compost or fertilizer.

3. Food Storage

Appropriate storage facilities for preserving food, controlling temperature and moisture, and minimizing spoilage.

4. Food Processing

A facility for preparing, processing, or packaging food while maintaining appropriate hygiene and sanitation standards.

5. Community Grocer / Food Hall

A publicly accessible area that supports food distribution, retail access, and community food services.

Teams must explain how these facilities operate individually and how they connect to form an efficient food security network.

4. FOUR DESIGN PILLARS

Pillar 1 - No Room for Error

Teams must demonstrate efficient land utilization, functional facility layouts, and effective transportation and pedestrian circulation throughout the district.

Designs should optimize food production, preservation, storage, and accessibility within the available land while considering safety, transportation efficiency, operational requirements, and opportunities for future expansion.

Pillar 2 - Mother Nature Meets Her Match

The proposed district must remain resilient during environmental disruptions, including flooding, freezing winters, and heat waves.

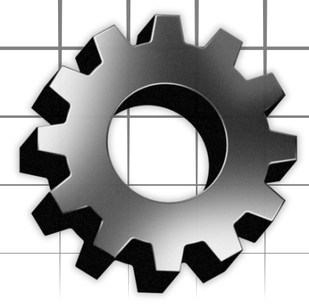
Teams should explain the engineering measures used to protect critical infrastructure and maintain essential operations.

Pillar 3 - Cost Beyond Construction

Teams must consider the long-term economic sustainability of their proposed district.

In addition to remaining within the $50 million CAD construction budget, teams must estimate annual operating, maintenance, and energy costs.

Teams should justify the long-term financial practicality of their design over its minimum 20-year lifespan, including reasonable estimates of recurring operating expenses, maintenance requirements,



energy consumption, and major equipment replacement needs.

Pillar 4 - Chain Reaction

Teams must consider how their proposed Urban Food Security District would maintain essential operations during unexpected supply-chain disruptions.

During the judging period, each team will receive a randomized supply-chain disruption scenario. Teams will have up to five (5) minutes to explain how their proposed district would respond to the disruption, identify the affected systems, and describe the operational adjustments or contingency measures required to maintain essential food production, preservation, and distribution.

Teams are encouraged to incorporate practical contingency measures into their original designs. However, they are not expected to predict the specific scenario they will receive.

The scenario will not be disclosed before the competition.

5. DESIGN CONSTRAINTS

Teams must satisfy the following land-use requirements within the 40 acre site:

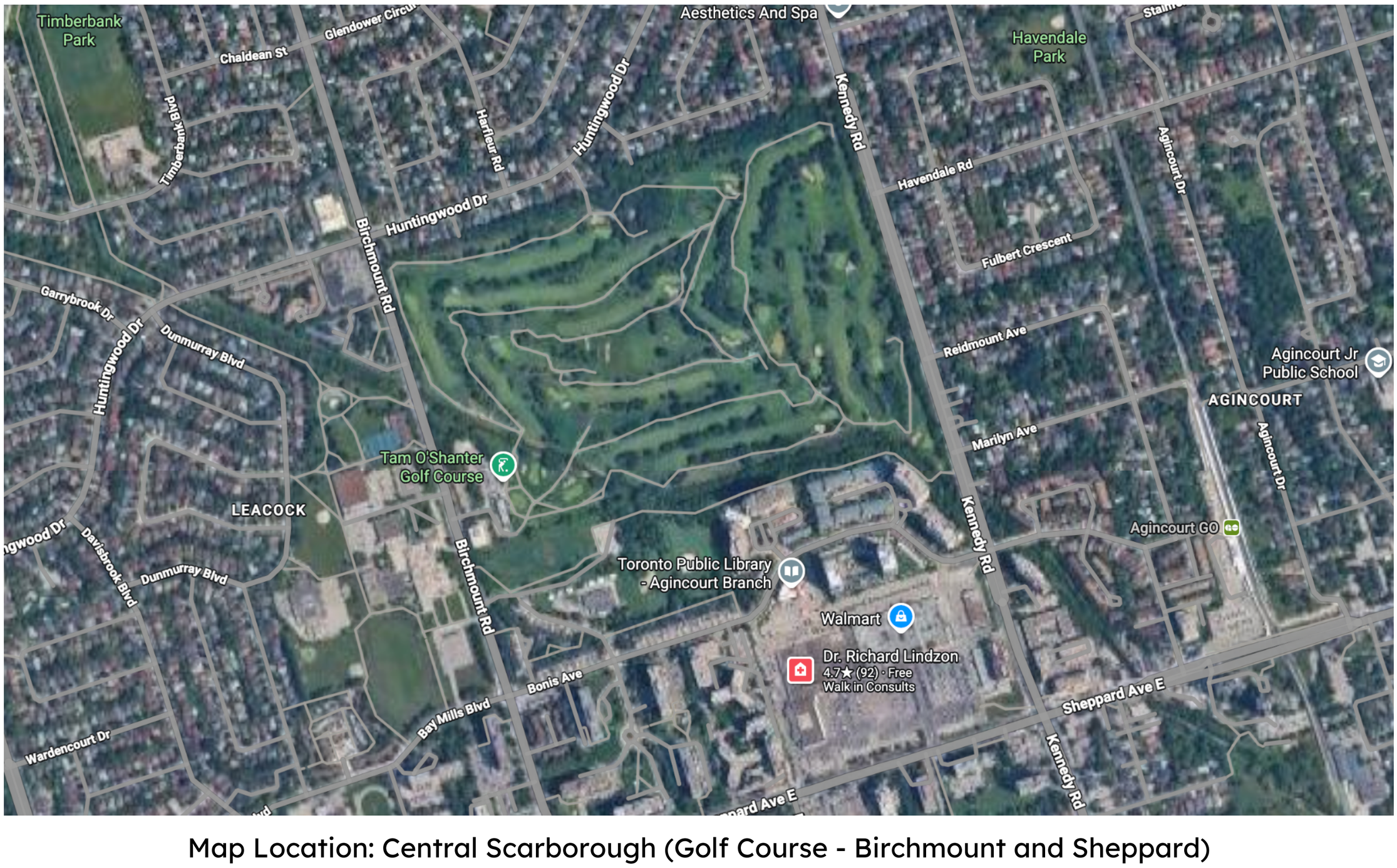
- A minimum of 20% of the total site must be allocated to open circulation and pedestrian-access areas.
- A minimum of 10% of the total site must be allocated to loading, transportation, and logistics operations.
- A minimum of 15% of the total site must be allocated to food storage facilities or designated storage operations.
- No individual facility or operational area may occupy more than 40% of the total site.

For the purposes of these minimum allocations, open circulation areas, loading and transportation areas, and food storage areas must be calculated separately and must not be double-counted.

Teams must provide a land-use breakdown showing the approximate area allocated to each major facility or operational function. The total allocated area must not exceed the available 40 acres.

The total initial construction and implementation budget must not exceed $50 million CAD. Teams must provide a reasonable cost breakdown covering the proposed facilities, infrastructure, equipment, and initial implementation requirements. Annual operating, maintenance, and energy costs must be estimated separately and considered when evaluating the district's long-term economic sustainability.

Teams must consider operational costs, maintenance costs, energy consumption, environmental resilience, and long-term accessibility.



Map Location: Central Scarborough (Golf Course - Birchmount and Sheppard)

6. DELIVERABLES

6.1 Design Summary

Teams must submit a maximum one-page Design Summary explaining:

- The proposed district and its primary objectives.
- The five required facilities and their functions.
- Major sustainability, accessibility, and financial considerations.
- The main engineering decisions and innovative features.
- How the proposed design addresses the four design pillars, including any planned contingency measures for supply-chain disruptions. Teams are not required to respond to the undisclosed Supply Chain Curveball scenario in their advance submission.

6.2 Presentation Slides

Teams must submit presentation slides in PDF format.

The presentation must include:

- An overall district site plan.
- Layouts or appropriate engineering representations of all five required facilities or operational areas, clearly identifying their functions, major equipment or systems, and connections to the overall district.
- Transportation and circulation routes within the district and surrounding community.
- An explanation of how the four design pillars were incorporated into the design.
- Engineering decisions and supporting justification.
- Construction costs and estimated annual operating expenses.
- Environmental resilience and sustainability considerations.

7. PRESENTATION AND JUDGING

Each team will have:

- 15 minutes for presentation.
- 5 minutes for judges' questions.
- Up to 5 minutes for the randomized Supply Chain Curveball.

All registered team members may contribute to the presentation, judges' questions, and Supply Chain Curveball response. Teams may briefly consult with one another during the curveball period, but the total response time must remain within the allotted five (5) minutes.

Teams will be evaluated on space efficiency, environmental resilience, economic sustainability, innovation, their Design Summary, and presentation performance.

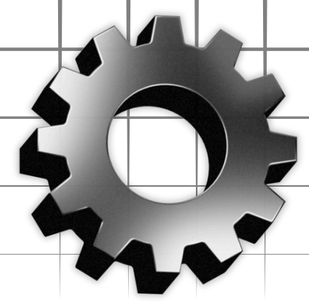
The complete judging rubric will be provided separately.

8. RULES AND GUIDELINES

Teams may use appropriate engineering software, research materials, calculations, and diagrams to support their proposed design.

All external sources must be properly cited.

The use of generative AI is not permitted for this competition. All research, ideas, designs, calculations, and submitted work must be completed by the registered team.



Teams must follow the official MEC 2026 competition rules and instructions provided by the Competition Coordinators.

Submission deadlines, check-in information, and presentation arrangements will be communicated through the official competition email at mec.vpcompetitions@mues.ca